Started on	Saturday, 31 January 2026, 10:02 AM
State	Finished
Completed on	Saturday, 31 January 2026, 11:00 AM
Time taken	57 mins 40 secs
Grade	Not yet graded

Question 1

Correct

Mark 1.00 out of 1.00

Given: A is 3×3,

$$(B^T - 2I)^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad |\,3A\,| = 27, \qquad \mathrm{tr}\,(3A) = 15.$$

Find $|\,A^{-1}\,|$

- A. 27
- B. 1 ✓
- C. $\frac{1}{3}$
- D. $\frac{1}{27}$

Your answer is correct.

The correct answer is: **1**

Question 2

Correct

Mark 1.00 out of 1.00

Given: A is 3×3,

$$(B^T - 2I)^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad |3A| = 27, \quad \text{tr}(3A) = 15.$$

Find B and B^2

- A. $B = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \ B^2 = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- B. $B = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \ B^2 = \begin{bmatrix} 25 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 9 \end{bmatrix}$ ✓
- C. $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \ B^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{bmatrix}$
- D. $B = \begin{bmatrix} 7 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 5 \end{bmatrix}, \ B^2 = \begin{bmatrix} 49 & 0 & 0 \\ 0 & 36 & 0 \\ 0 & 0 & 25 \end{bmatrix}$

Your answer is correct.

The correct answer is: $B = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \ B^2 = \begin{bmatrix} 25 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 9 \end{bmatrix}$

Question 3

Correct

Mark 1.00 out of 1.00

Given: A is 3×3,

$$(B^T - 2I)^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad |3A| = 27, \quad \text{tr}(3A) = 15.$$

Find $|2AB|$

- A. 60
- B. 240
- C. 480 ✓
- D. 960

Your answer is correct.

The correct answer is: 480

Question 4

Correct

Mark 1.00 out of 1.00

Given: A is 3×3,

$$(B^T - 2I)^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad |3A| = 27, \qquad \text{tr}(3A) = 15.$$

Find $\text{tr}(2A + 3B)$

- A. 36
- B. 46 ✓
- C. 56
- D. 26

Your answer is correct.

The correct answer is: 46

Question 5

Correct

Mark 1.00 out of 1.00

Encode the message " ***Help him now*** " using the given table and the matrix, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

_ = 0	I = 9	R = 18
A = 1	J = 10	S = 19
B = 2	K = 11	T = 20
C = 3	L = 12	U = 21
D = 4	M = 13	V = 22
E = 5	N = 14	W = 23
F = 6	O = 15	X = 24
G = 7	P = 16	Y = 25
H = 8	Q = 17	Z = 26

Choose the correct encoded sequence:

- A. $\mathbf{13, 5, 28, 16, 8, 8, 22, 13, 14, 14, 38, 23}$ ✓
- B. $\mathbf{13, 28, 8, 22, 14, 38, 5, 16, 8, 13, 14, 23}$
- C. $\mathbf{8, 5, 12, 16, 0, 8, 9, 13, 0, 14, 15, 23}$
- D. $\mathbf{5, 13, 16, 28, 8, 8, 13, 22, 14, 14, 23, 38}$

Your answer is correct.

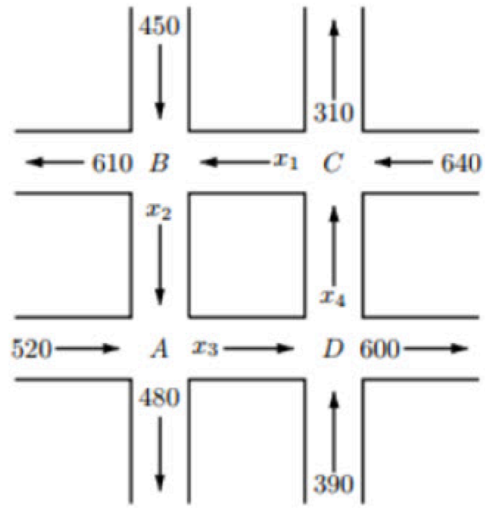
The correct answer is: $\mathbf{13, 5, 28, 16, 8, 8, 22, 13, 14, 14, 38, 23}$

Question 6

Correct

Mark 1.00 out of 1.00

In an AI-based traffic digital twin, the unknown internal one-way flows are x_1, x_2, x_3, x_4. Using flow conservation at each junction (Input = Output), The governing linear system is:



Which of the following systems matches the conservation equations for the network?

- A. $\begin{cases} x_2 + 520 = x_3 + 480 \\ x_1 + 450 = x_2 + 610 \\ x_4 + 640 = x_1 + 310 \\ x_3 + 390 = x_4 + 600 \end{cases}$ ✓
- B. $\begin{cases} x_3 + 520 = x_2 + 480 \\ x_2 + 450 = x_1 + 610 \\ x_1 + 640 = x_4 + 310 \\ x_4 + 390 = x_3 + 600 \end{cases}$
- C. $\begin{cases} x_2 + 480 = x_3 + 520 \\ x_1 + 610 = x_2 + 450 \\ x_4 + 310 = x_1 + 640 \\ x_3 + 600 = x_4 + 390 \end{cases}$
- D. $\begin{cases} x_2 - 520 = x_3 - 480 \\ x_1 - 450 = x_2 - 610 \\ x_4 - 640 = x_1 - 310 \\ x_3 - 390 = x_4 - 600 \end{cases}$

Your answer is correct.

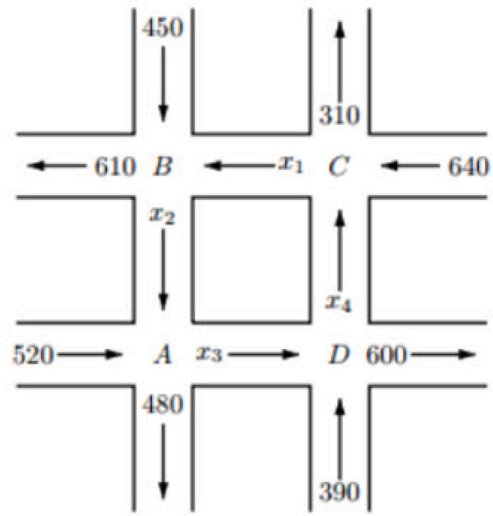
The correct answer is: $\begin{cases} x_2 + 520 = x_3 + 480 \\ x_1 + 450 = x_2 + 610 \\ x_4 + 640 = x_1 + 310 \\ x_3 + 390 = x_4 + 600 \end{cases}$

Question 7

Correct

Mark 1.00 out of 1.00

In an AI-based traffic digital twin, the unknown internal one-way flows are x_1, x_2, x_3, x_4. Using flow conservation at each junction (Input = Output), The governing linear system is:



For the same traffic model, which statement is correct?

- A. **The system has a unique solution:** $(x_1, x_2, x_3, x_4) = (330, 170, 210, 0)$
- B. **The system has infinitely many solutions:** $(x_1, x_2, x_3, x_4) = (t + 330,\ t + 170,\ t + 210,\ t), t \geq 0$ ✓
- C. **The system has no solution because the equations are inconsistent.**
- D. **The system has infinitely many solutions:** $(x_1, x_2, x_3, x_4) = (t,\ t + 330,\ t + 170,\ t + 210), t \geq 0$

Your answer is correct.

The correct answer is:
The system has infinitely many solutions:
$(x_1, x_2, x_3, x_4) = (t + 330,\ t + 170,\ t + 210,\ t), t \geq 0$

Question 8
Correct
Mark 1.00 out of 1.00

In the traffic digital twin model, the internal flows satisfy the general solution:
$(x_1, x_2, x_3, x_4) = (t + 330,\ t + 170,\ t + 210,\ t)$
where t is a free parameter.
Assume two practical AI constraints from sensors/road capacity:

- **Nonnegativity:** $x_i \geq 0$ **for all** i
- **Capacity constraint:** $x_1 \leq 500$

Which range of t makes the solution physically feasible?

- A. $0 \leq t \leq 500$
- B. $-330 \leq t \leq 170$
- C. $0 \leq t \leq 170$ ✓
- D. $t \geq 0$ **only**

Your answer is correct.

The correct answer is: $0 \leq t \leq 170$

Question 9
Incorrect
Mark 0.00 out of 1.00

In AI/Data Science, eigenvectors define "special directions" that are scaled (not rotated) by a linear transform A.
Let A be an $n \times n$ matrix and $v \neq 0$ be an eigenvector of A with eigenvalue λ, i.e. $Av = \lambda v$.

Which statement is always true?

- A. v **is also an eigenvector of** A^2 **with eigenvalue** λ^2.
- B. v **is always an eigenvector of** A^{-1} **with eigenvalue** λ.
- C. **If** v **is an eigenvector of** A, **then** v **must be an eigenvector of** A^T. ✗
- D. **If** $Av = \lambda v$, **then** v **is also an eigenvector of** $A + I$, **with eigenvalue** λ.

Your answer is incorrect.

The correct answer is: v **is also an eigenvector of** A^2 **with eigenvalue** λ^2.

Question 10

Correct

Mark 1.00 out of 1.00

In signal processing and dynamical models used in AI, matrix functions like $\cos(A)$ are defined using power series and can be computed efficiently if A is diagonalizable.

Assume A is diagonalizable:

$$A = PDP^{-1}, \qquad D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Which expression is correct for $\cos(A)$?

- A. $\cos(A) = P\cos(D)P^{-1}$, where $\cos(D) = \text{diag}(\cos\lambda_1, \dots, \cos\lambda_n)$ ✓
- B. $\cos(A) = P^{-1}\cos(D)P$, where $\cos(D) = \text{diag}(\cos\lambda_1, \dots, \cos\lambda_n)$
- C. $\cos(A) = \cos(P)\cos(D)\cos(P^{-1})$
- D. $\cos(A) = \text{diag}(\cos\lambda_1, \dots, \cos\lambda_n)$ always, even if A is not diagonal

Your answer is correct.

The correct answer is: $\cos(A) = P\cos(D)P^{-1}$, where $\cos(D) = \text{diag}(\cos\lambda_1, \dots, \cos\lambda_n)$

Question 11

Correct

Mark 1.00 out of 1.00

If $A = \begin{bmatrix} 3 & -3 & -4 \\ 0 & 2 & 2 \\ 0 & 0 & k \end{bmatrix}$, and $|A| = -6$.

i) Find the eigenvalues of the matrices A, $2A$, A^T, A^3, A^{-1}.

Which option is completely correct?

A. $k = -1$.
Eigenvalues:
- $A: \{-1, 2, 3\}$
- $2A: \{-2, 4, 6\}$
- $A^T: \{-1, 2, 3\}$
- $A^3: \{-1, 8, 27\}$
- $A^{-1}: \left\{-1, \frac{1}{2}, \frac{1}{3}\right\}$

✓

B. $k = -1$.
Eigenvalues:
- $A: \{-1, 2, 3\}$
- $2A: \{-1, 4, 6\}$
- $A^T: \{-1, 2, 3\}$
- $A^3: \{-1, 8, 27\}$
- $A^{-1}: \left\{-1, \frac{1}{2}, \frac{1}{3}\right\}$

C. $k = 1$.
Eigenvalues:
- $A: \{1, 2, 3\}$
- $2A: \{2, 4, 6\}$
- $A^T: \{1, 2, 3\}$
- $A^3: \{1, 8, 27\}$
- $A^{-1}: \left\{1, \frac{1}{2}, \frac{1}{3}\right\}$

D. $k = -2$.
Eigenvalues:
- $A: \{-2, 2, 3\}$
- $2A: \{-4, 4, 6\}$
- $A^T: \{-2, 2, 3\}$
- $A^3: \{-8, 8, 27\}$
- $A^{-1}: \left\{-\frac{1}{2}, \frac{1}{2}, \frac{1}{3}\right\}$

Your answer is correct.

The correct answer is:

$k = -1$.

Eigenvalues:

- $A: \{-1, 2, 3\}$
- $2A: \{-2, 4, 6\}$
- $A^T: \{-1, 2, 3\}$
- $A^3: \{-1, 8, 27\}$
- $A^{-1}: \left\{-1, \frac{1}{2}, \frac{1}{3}\right\}$

Question **12**

Correct

Mark 2.00 out of 2.00

If $A = \begin{bmatrix} 3 & -3 & -4 \\ 0 & 2 & 2 \\ 0 & 0 & k \end{bmatrix}$, and $|A| = -6.$

ii) Find the eigenvectors of the matrix A.

Which option lists three vectors that are all eigenvectors of A?

A. $\left\{\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\0\\0\end{bmatrix}\right\}$

B. $\left\{\begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}\right\}$

C. $\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$

D. $\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}3\\1\\0\end{bmatrix}, \begin{bmatrix}3\\-4\\6\end{bmatrix}\right\}$ ✓

Your answer is correct.

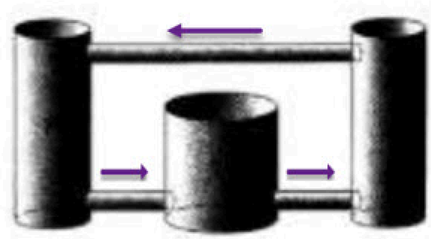
The correct answer is: $\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}3\\1\\0\end{bmatrix}, \begin{bmatrix}3\\-4\\6\end{bmatrix}\right\}$

Question 13

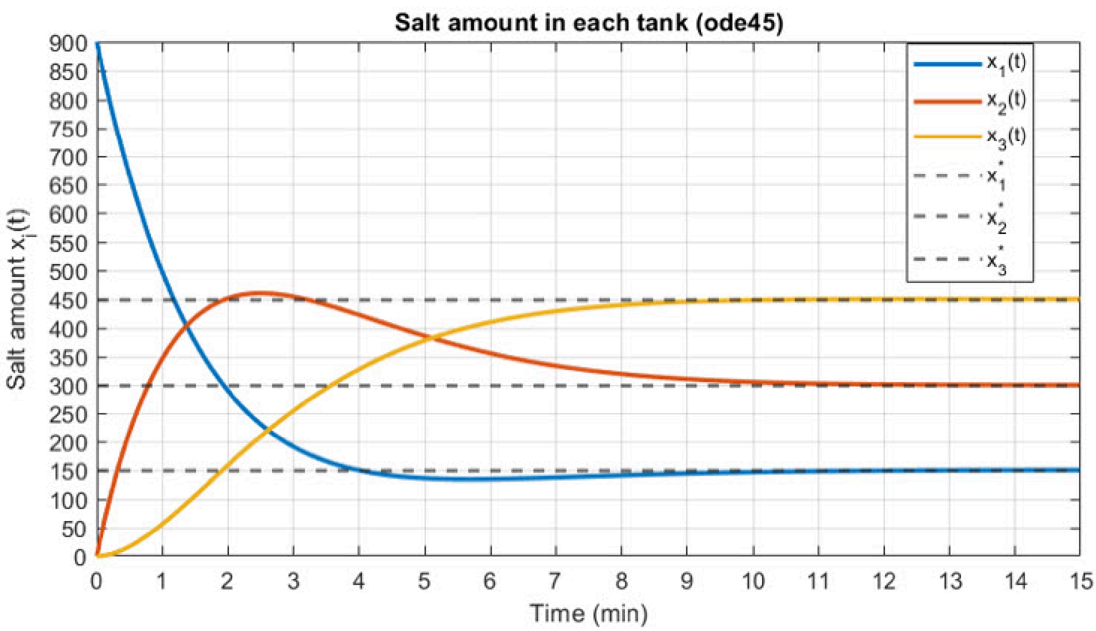
Complete

Marked out of 2.00

A closed loop consists of three well-mixed tanks connected by one-way pipes. Brine flows from Tank 1 → Tank 2 → Tank 3 → back to Tank 1 with a constant flow rate r. The tanks have volumes V_1, V_2, V_3. Let $x_1(t), x_2(t), x_3(t)$ denote the amount of salt in Tanks 1, 2, and 3 (respectively) at time t(minutes). You are provided with:



1. **The process description + diagram of the three-tank loop.**
2. **The time-response plot of $x_1(t), x_2(t), x_3(t)$(salt amount in each tank versus time), as shown.**



Required to answer

a) Using the plot, **and what happens during the first stage of the process** (near $t = 0$):

- Which tank initially has the largest salt amount and why? With stating the initial amount of Salt.
- Which tank initially has zero or very small salt amount and why?

b) Using **the plot and the "well-mixed, constant-flow" assumptions, deduce a Relationship between tank volumes i.e. (V_1: V_2: V_3).**

c) The Amount of Salt in Each Tank after 180 second and 720 second.

a) The Tank **X1** has the **largest** salt with initial amount of salt **900**

The Tank **X2** has a very small salt amount & The Tank **X3** has Zero

b) X1 = - r/V1 X1 + r/V2 X2

X2 = - r/V1 X2 + r/V2 X3

c) The Amount of salt in each tank after 180 sec >> **X1 = 200 , X2 = 450 , X3 = 250**

The Amount of salt in each tank after 720 sec >> **X1 = 150 , X2 = 300 , X3 = 450**